# Interpretable Machine Learning

# Leave One Covariate Out (LOCO)

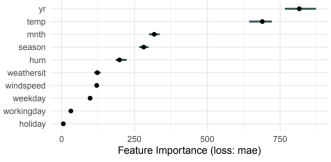


**Figure:** Bike Sharing Dataset

## Learning goals

- Definition of LOCO
- Interpretation of LOCO

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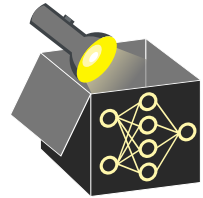
Lei et al. (2018)

Tibshirani (2018)



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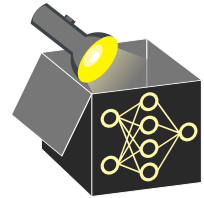

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**Definition:** Given training and test datasets $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}} \subseteq \mathcal{D}$, some $\mathcal{I}$ and a model $\hat{f} = \mathcal{I}(\mathcal{D}_{\text{train}})$. Then LOCO for a feature $j \in \{1, \ldots, p\}$ can be computed as follows:

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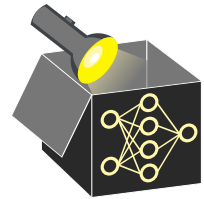

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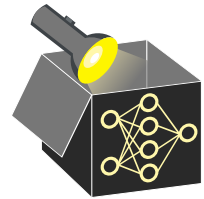

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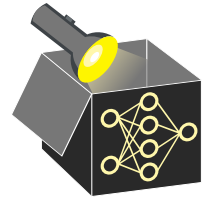

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The method can be generalized to other loss functions and aggregations. If we use mean instead of median we can rewrite LOCO as

$$\text{LOCO}_j = \mathcal{R}_{\text{emp}}(\hat{f}_{-j}) - \mathcal{R}_{\text{emp}}(\hat{f}).$$

# BIKE SHARING EXAMPLE

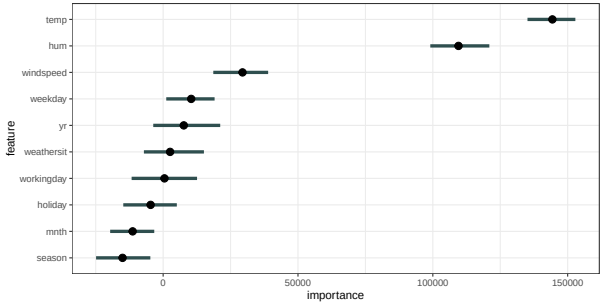
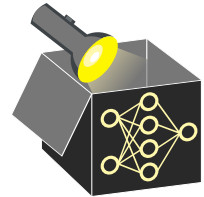



**Figure:** A random forest with default hyperparameters was fit on 70% of the bike sharing data (training set) to optimize MSE. Then LOCO was computed for all features on the test data. The temperature is the most important feature. Without access to $\mathrm{temp}$, the MSE increases by approx. $140,000$.

# INTERPRETATION OF LOCO



**Interpretation:** LOCO estimates the generalization error of the learner on a reduced dataset $\mathcal{D}_{-j}$.

Can we get insight into whether the ...

1. feature $x_j$ is causal for the prediction $\hat{y}$?
   - In general, no also because we refit the model (counterexample on the next slide)
2. feature $x_j$ contains prediction-relevant information?
   - In general, no (counterexample on the next slide)
3. model requires access to $x_j$ to achieve its prediction performance?
   - Approximately, it provides insight into whether the *learner* requires access to $x_j$

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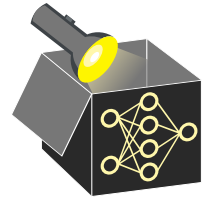

**Example:** Sample 1000 observations with

- $x_1, x_3 \sim N(0, 5)$
- $x_2 = x_1 + \epsilon_2$ with $\epsilon_2 \sim N(0, 0.1)$
- $y = x_2 + x_3 + \epsilon$ with $\epsilon \sim N(0, 2)$

$\Rightarrow$ Fitting a LM yields
$\hat{f}(x) = -0.02 - 1.02x_1 + 2.05x_2 + 0.98x_3$

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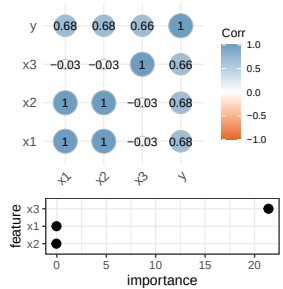


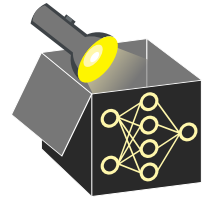

Top: Correlation matrix
Bottom: LOCO importance of LM fitted on 70% of the data computed on 30% remaining observations

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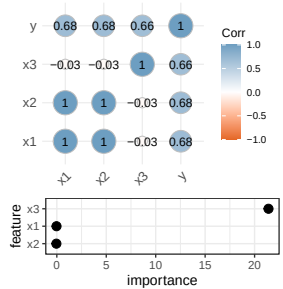


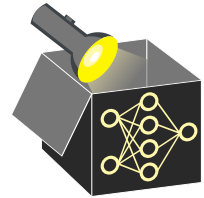

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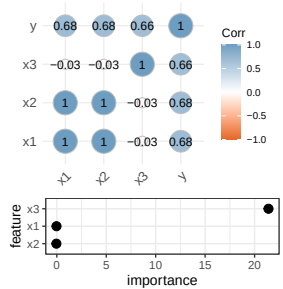


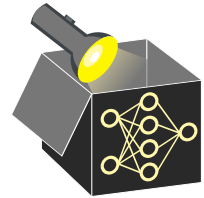

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$\Rightarrow$ We also can't infer (2), e.g., $Cor(x_2, y) = 0.68$ but $\text{LOCO}_2 \approx 0$

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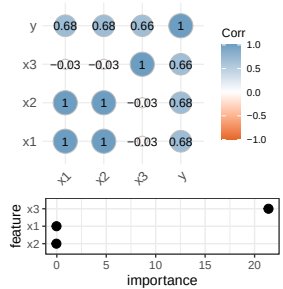


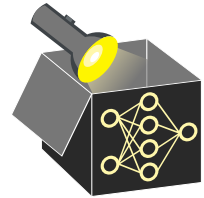

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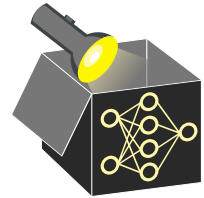
$\Rightarrow$ We also can't infer (2), e.g., $Cor(x_2, y) = 0.68$ but $\text{LOCO}_2 \approx 0$
$\Rightarrow$ We can get insight into (3): $x_2$ and $x_1$ highly correlated with $\text{LOCO}_1 = \text{LOCO}_2 \approx 0$
$\rightsquigarrow$ $x_2$ and $x_1$ take each others place if one of them is left out (not the case for $x_3$)

# PROS AND CONS



Pros:

- Requires (only?) one refitting step per feature for evaluation
- Easy to implement
- Testing framework available in ▸ Lei et al. (2018)

Cons:

- Does not provide insight into a specific model, but rather a learner on a specific dataset
- Model training is a random process, so estimates can be noisy (which is problematic for inference about model and data)
- Requires re-fitting the learner for each feature $\rightarrow$ computationally intensive compared to PFI